

Tasks

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Answer :

Step1 : Open Command Prompt and run the command → `ipconfig /all`

IPv4 Address. : 192.168.182.141

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.182.2

DNS Servers : 192.168.182.2

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Answer :

```
C:\Users\cdiks>ping www.google.com

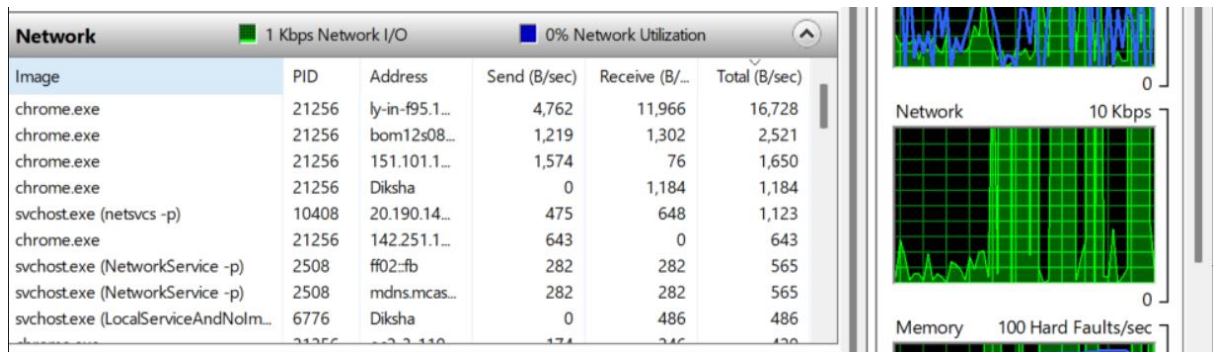
Pinging www.google.com [142.251.153.119] with 32 bytes of data:
Reply from 142.251.153.119: bytes=32 time=15ms TTL=128
Reply from 142.251.153.119: bytes=32 time=14ms TTL=128
Reply from 142.251.153.119: bytes=32 time=19ms TTL=128
Reply from 142.251.153.119: bytes=32 time=18ms TTL=128

Ping statistics for 142.251.153.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 19ms, Average = 16ms
```

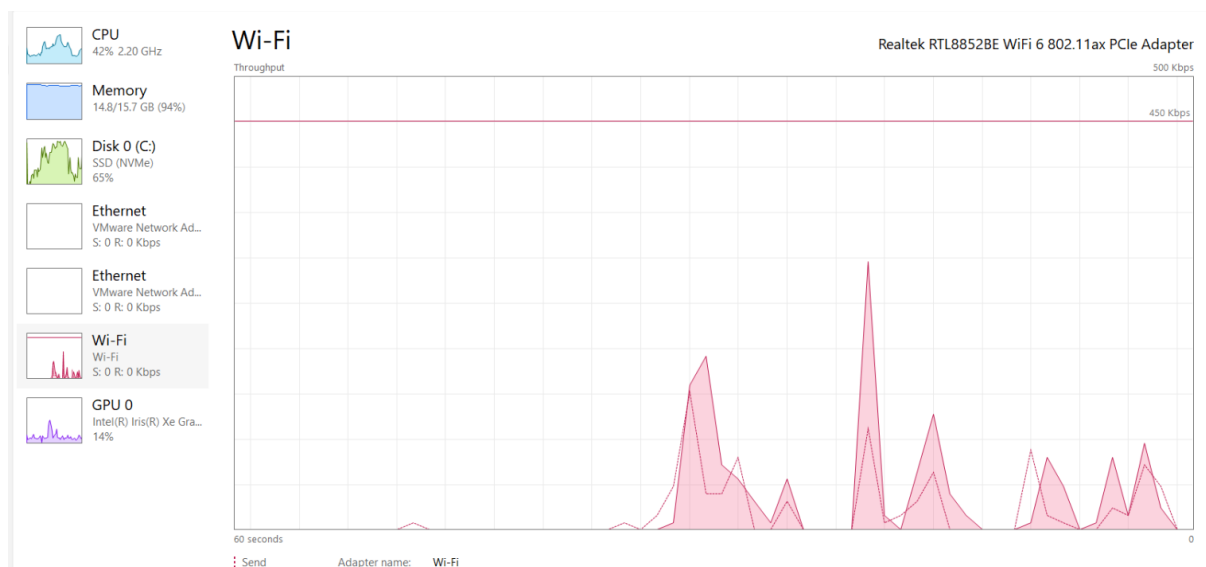
3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Answer :

Resource Monitor



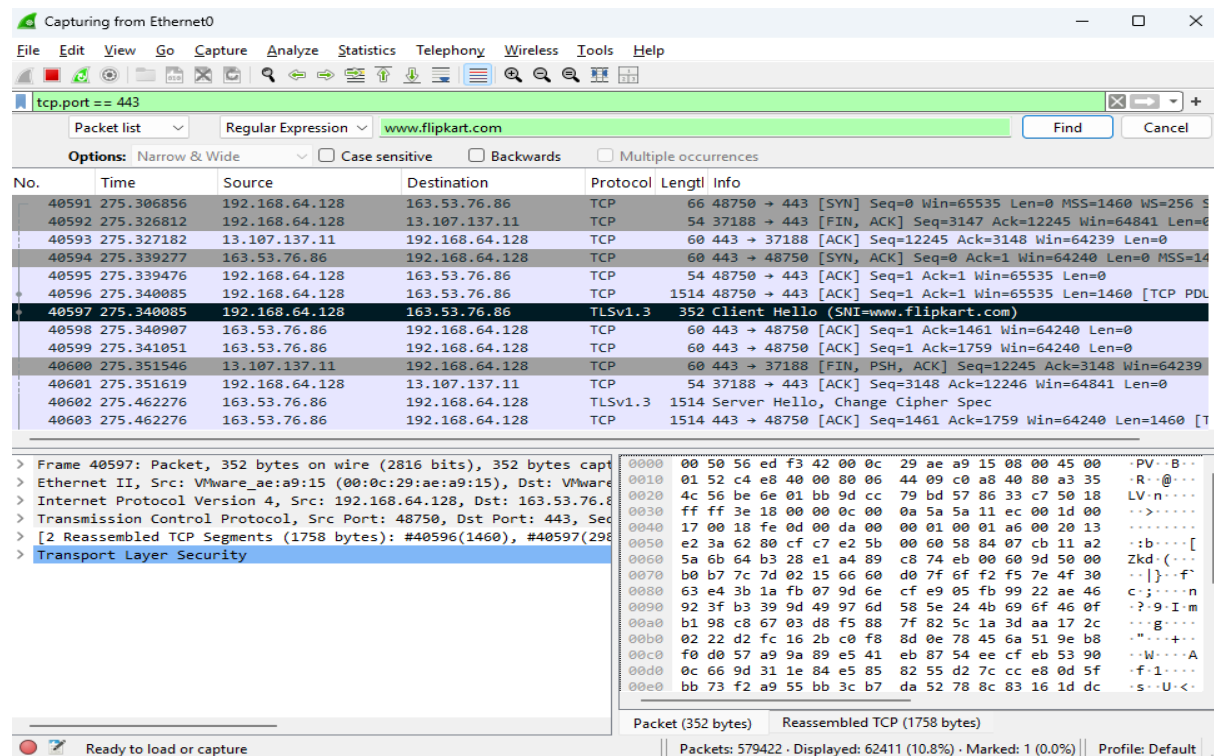
Task Manager



I checked the network usage for 5 minutes while watching a YouTube video. The highest network usage was about 16.7 KB/s. Google Chrome (chrome.exe) used the most internet because the video was playing in Chrome.

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Answer :



5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Answer : I used ChatGPT to generate the following troubleshooting checklist for improving internet speed while gaming:

1. Restart the Wi-Fi router.
2. Use an Ethernet cable instead of Wi-Fi.
3. Close background apps and downloads.
4. Move closer to the router.
5. Check internet speed and ping regularly.

Suggestion I would try first: Restart the Wi-Fi router because it is simple and often fixes internet problems quickly.